

Endsemester Solutions with Marking Scheme

EE 708 (Information Theory & Coding) Q1–Q3, Q9–Q10

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Question 1

(2 + 2 + 2 = 6 marks)

Setup. $X_0 \sim \text{Unif}\{1, \dots, 24\}$, $X_1 \sim \text{Unif}\{9, \dots, 32\}$, $\Theta \sim \text{Bernoulli}(1/2)$ (independent of X_0, X_1). Define $X := X_\Theta$.

The alphabets overlap on $\{9, \dots, 24\}$ (16 values). Non-overlap regions: $\{1, \dots, 8\}$ and $\{25, \dots, 32\}$ (8 values each).

Part (i): $H(X)$

[2]

By total probability,

$$\begin{aligned} P(x) &= \sum_{\theta=0}^1 P(X = x | \Theta = \theta) P(\Theta = \theta) \\ &= \frac{1}{2} P(X = x | \Theta = 0) + \frac{1}{2} P(X = x | \Theta = 1) \\ &= \frac{1}{2} P(X_0 = x) + \frac{1}{2} P(X_1 = x) \\ &= \begin{cases} \frac{1}{48}, & x \in \{1, \dots, 8\} \cup \{25, \dots, 32\}, \\ \frac{1}{24}, & x \in \{9, \dots, 24\}. \end{cases} \end{aligned}$$

$$H(X) = 16 \cdot \frac{1}{48} \log 48 + 16 \cdot \frac{1}{24} \log 24 = \frac{1}{3}(\log 24 + 1) + \frac{2}{3} \log 24 = \log_2 24 + \frac{1}{3}.$$

$$H(X) = \log_2 24 + \frac{1}{3} = \log_2 3 + \frac{10}{3} \approx 4.918 \text{ bits.}$$

Marking Scheme

Correct PMF of X (non-overlap = $\frac{1}{48}$, overlap = $\frac{1}{24}$)	1
Correct entropy computation, simplified value	1
Subtotal	2

Part (ii): $H(X | \Theta)$

[2]

Given $\Theta = \theta$, $X = X_\theta$ is uniform on a 24-element set, so $H(X | \Theta = \theta) = \log_2 24$ for both θ .

$$H(X | \Theta) = \frac{1}{2} \log_2 24 + \frac{1}{2} \log_2 24 = \log_2 24 = 3 + \log_2 3.$$

$$H(X | \Theta) = 3 + \log_2 3 \approx 4.585 \text{ bits.}$$

Marking Scheme

Identifying that $X \Theta$ is uniform on a 24-element set	1
Correct value $3 + \log_2 3$	1
Subtotal	2

Part (iii): $H(\Theta | X)$ [2]

By chain rule, $H(\Theta | X) = H(\Theta) + H(X | \Theta) - H(X) = 1 + \log 24 - (\log 24 + 1/3) = 2/3$.

$$H(\Theta | X) = \frac{2}{3} \text{ bit.}$$

Marking Scheme

Correct chain rule formula	1
Correct final value $2/3$	1
Subtotal	2

Question 2 (3 marks)

Claim. $S := \{p_X : I(X; Y) = \max\}$ is convex.

Step 1 – Concavity of $I(X; Y)$ in p_X . Fix the channel $p_{Y|X}$. For inputs p_1, p_2 and $\lambda \in [0, 1]$, we know

$$I_{\lambda p_1 + (1-\lambda)p_2}(X; Y) \geq \lambda I_{p_1}(X; Y) + (1-\lambda)I_{p_2}(X; Y).$$

This shows concavity of $I(X; Y)$ in p_X .

Step 2 – Convexity of S . Let $C := \max_{p_X} I(X; Y)$. Take $p_1, p_2 \in S$. So, $I_{p_1}(X; Y) = I_{p_2}(X; Y) = C$. For $\lambda \in [0, 1]$, $\lambda p_1 + (1-\lambda)p_2$ is a valid pmf

By concavity of $I(X; Y)$,

$$I_{\lambda p_1 + (1-\lambda)p_2}(X; Y) \geq \lambda C + (1-\lambda)C = C.$$

But C is the maximum, so equality must hold and $\lambda p_1 + (1-\lambda)p_2 \in S$. Hence S is convex. □

Marking Scheme

Statement of concavity of $I(X; Y)$ in p_X	1
Statement of $\lambda p_1 + (1-\lambda)p_2$ being a valid pmf	1
Concluding convexity of S from concavity + maximization	1
Subtotal	3

Question 3 (2 + 2 + 2 + 2 = 8 marks)

Part (a): $I(X; Y | Z) \leq I(X; Y)$? [2]

False. *Counter-example.* Let $X, Y \stackrel{\text{iid}}{\sim} \text{Bernoulli}(1/2)$ and $Z = X \oplus Y$.

- $X \perp Y \Rightarrow I(X; Y) = 0$.
- Given Z , $Y = X \oplus Z$ is determined by (X, Z) , so $H(Y | X, Z) = 0$, while $Y | Z \sim \text{Bernoulli}(1/2)$. Hence $I(X; Y | Z) = H(Y | Z) - H(Y | X, Z) = 1$ bit.

So $I(X; Y | Z) = 1 > 0 = I(X; Y)$.

Marking Scheme

Stating “False”	1
Valid counter-example with both $I(X; Y)$ and $I(X; Y Z)$ correctly computed	1
Subtotal	2

Part (b): $D(p||q) = D(q||p)$? [2]

False. *Counter-example.* Let $p = (1/2, 1/2)$ and $q = (0, 1)$ on $\{0, 1\}$.

$$D(p||q) = \frac{1}{2} \log \frac{1/2}{0} + \frac{1}{2} \log \frac{1/2}{1} = \text{inf. bits},$$

$$D(q||p) = 0 \log \frac{0}{1/2} + \log \frac{1}{1/2} = 0 + 1 = 1 \text{ bit}.$$

Since $D(p||q) \neq D(q||p)$, the claim fails.

Marking Scheme

Stating “False”	1
Valid counter-example with both KL values explicitly computed	1
Subtotal	2

Part (c): $h(aX) \neq h(X)$ for any continuous X and $a \neq \pm 1$? [2]

True (for real $a \notin \{-1, 0, 1\}$; the case $a = 0$ is degenerate).

Justification. For any nonzero real a and continuous X with density,

$$h(aX) = h(X) + \log |a|.$$

For $a \neq \pm 1$ (and nonzero), $|a| \neq 1$, so $\log |a| \neq 0$, hence $h(aX) \neq h(X)$. The cases $a = \pm 1$ are precisely the ones excluded because $|-1| = |1| = 1$ gives equality.

Marking Scheme

Stating the scaling identity $h(aX) = h(X) + \log a $	1
Concluding “True” for $a \neq \pm 1$ and addressing $ a \neq 1 \Rightarrow \log a \neq 0$	1
Subtotal	2

Part (d): Unique decodability $\Rightarrow$ instantaneous decoding? [2]

False. Unique decodability is strictly weaker than instantaneous (prefix-free) decodability.

Counter-example. Code $\{a \rightarrow 0, b \rightarrow 01, c \rightarrow 011\}$.

- *Uniquely decodable*: Every codeword starts with 0. The number of 1s following each 0 identifies the symbol. Decoding is unambiguous.
- *Not instantaneous*: 0 is a prefix of 01, which is a prefix of 011. After receiving a 0, the decoder must look at *future* bits to decide which symbol was transmitted.

Marking Scheme

Stating “False” and distinguishing UD vs prefix-free	1
Concrete counter-example (e.g. $\{0, 01, 011\}$ or $\{10, 00, 11, 110\}$) with brief explanation	1
Subtotal	2

Question 9

(3 marks)

Differential entropy of the Laplacian $f_Z(z) = \frac{1}{2b}e^{-|z|/b}$.

Step 1 – $\mathbb{E}[|Z|]$. By symmetry,

$$\mathbb{E}[|Z|] = 2 \int_0^\infty z \cdot \frac{1}{2b}e^{-z/b} dz = \frac{1}{b} \int_0^\infty z e^{-z/b} dz = \frac{1}{b} \cdot b^2 = b.$$

Step 2 – **Differential entropy**. Using $\ln f_Z(z) = -\ln(2b) - |z|/b$ (natural log),

$$h(Z) = -\mathbb{E}[\ln f_Z(Z)] = \ln(2b) + \frac{\mathbb{E}[|Z|]}{b} = \ln(2b) + 1 = \ln(2eb) \text{ nats.}$$

Convert to bits: $h(Z) = \log_2(2eb)$ bits.

$$h(Z) = \ln(2eb) \text{ nats} = \log_2(2eb) \text{ bits.}$$

Marking Scheme

Setting up $h(Z) = -\mathbb{E}[\ln f_Z(Z)] = \ln(2b) + \mathbb{E}[Z]/b$	1
Correctly evaluating $\mathbb{E}[Z] = b$	1.5
Correct final expression $\ln(2eb)$ (or $\log_2(2eb)$)	0.5
Subtotal	3

Question 10

(4 marks)

Claim. For the channel $Y = X + Z$ with $X \perp Z$, $\mathbb{E}X^2 \leq P$, the capacity

$$C(Z) = \max_{p_X: \mathbb{E}X^2 \leq P} I(X; Y)$$

is minimized over $\{Z : \mathbb{E}Z^2 \leq \sigma^2\}$ by $Z^* \sim \mathcal{N}(0, \sigma^2)$, with minimum value $\frac{1}{2} \log(1 + P/\sigma^2)$.

Tools. (T1) Entropy Power Inequality: for $X \perp Z$, $2^{2h(X+Z)} \geq 2^{2h(X)} + 2^{2h(Z)}$, with equality iff both Gaussian.

(T2) Gaussian maximizes differential entropy under a variance constraint: $h(W) \leq \frac{1}{2} \log(2\pi e \mathbb{E}W^2)$, with equality iff W is Gaussian.

Step 1 – Simplify $I(X; Y)$. Since $X \perp Z$,

$$h(Y | X) = h(X + Z | X) = h(Z | X) = h(Z),$$

so $I(X; Y) = h(Y) - h(Z) = h(X + Z) - h(Z)$ and

$$C(Z) = \max_{\mathbb{E}X^2 \leq P} [h(X + Z) - h(Z)].$$

Step 2 – Lower-bound $h(X + Z)$ via EPI. Apply (T1):

$$h(X + Z) \geq \frac{1}{2} \log(2^{2h(X)} + 2^{2h(Z)}).$$

Substituting into the max,

$$C(Z) \geq \max_{\mathbb{E}X^2 \leq P} \frac{1}{2} \log(2^{2h(X)} + 2^{2h(Z)}) - h(Z).$$

Step 3 – Push the max inside. $\log$ is monotone increasing, $2^{2h(Z)}$ does not depend on X , so

$$\max_{\mathbb{E}X^2 \leq P} \frac{1}{2} \log(2^{2h(X)} + 2^{2h(Z)}) = \frac{1}{2} \log(2^{2 \max_{\mathbb{E}X^2 \leq P} h(X)} + 2^{2h(Z)}).$$

By (T2), $\max_{\mathbb{E}X^2 \leq P} h(X) = \frac{1}{2} \log(2\pi e P)$, attained by $X \sim \mathcal{N}(0, P)$. Thus $2^{2 \max_{\mathbb{E}X^2 \leq P} h(X)} = 2\pi e P$ and

$$C(Z) \geq \frac{1}{2} \log(2\pi e P + 2^{2h(Z)}) - h(Z) = \frac{1}{2} \log\left(\frac{2\pi e P}{2^{2h(Z)}} + 1\right).$$

Step 4 – Use the variance constraint on Z . By (T2), $h(Z) \leq \frac{1}{2} \log(2\pi e \mathbb{E}Z^2) \leq \frac{1}{2} \log(2\pi e \sigma^2)$, hence $2^{2h(Z)} \leq 2\pi e \sigma^2$. Substituting,

$$C(Z) \geq \frac{1}{2} \log\left(\frac{2\pi e P}{2\pi e \sigma^2} + 1\right) = \frac{1}{2} \log\left(1 + \frac{P}{\sigma^2}\right).$$

Step 5 – Achievability at $Z^* \sim \mathcal{N}(0, \sigma^2)$. When Z^* is Gaussian, choosing $X^* \sim \mathcal{N}(0, P)$ makes both X^* and Z^* Gaussian:

- EPI is tight: $2^{2h(X^*+Z^*)} = 2\pi e P + 2\pi e \sigma^2 = 2^{2h(X^*)} + 2^{2h(Z^*)}$.
- Variance-entropy bound is tight: $h(Z^*) = \frac{1}{2} \log(2\pi e \sigma^2)$.

Hence $C(Z^*) = \frac{1}{2} \log(1 + P/\sigma^2)$ exactly.

Combining Steps 4 and 5:

$$C(Z) \geq \frac{1}{2} \log(1 + P/\sigma^2) = C(Z^*) \quad \text{for every admissible } Z,$$

so Gaussian noise of variance σ^2 minimizes the capacity. □

$$\min_{Z: \mathbb{E}Z^2 \leq \sigma^2} C(Z) = \frac{1}{2} \log\left(1 + \frac{P}{\sigma^2}\right), \text{ attained by } Z^* \sim \mathcal{N}(0, \sigma^2).$$

Among all additive-noise channels with a fixed noise-variance budget, AWGN has the minimum capacity.

Marking Scheme

Reducing $I(X; Y)$ to $h(X + Z) - h(Z)$ using $h(Y X) = h(Z)$	0.5
Applying EPI inside the max to lower-bound $h(X + Z)$	1
Pushing the max through log and using $\max_{\mathbb{E}X^2 \leq P} h(X) = \frac{1}{2} \log(2\pi eP)$	1
Using $h(Z) \leq \frac{1}{2} \log(2\pi e\sigma^2)$ to obtain the final bound $\frac{1}{2} \log(1 + P/\sigma^2)$	1
Achievability: verifying both EPI and variance-entropy bound are tight at Z^*	0.5
Subtotal	4

Appropriate partial marks will be awarded to solutions based on the level of completeness and correctness.

Q4:

B) best describes **Optimal codes**. Any other option or multiple options stated as correct are marked a 0.

Q5:

- a) **0.5 marks** each for calculating $H(X)$ and $H(Y)$ correctly.
- b) There are two methods, one by first principles approach and another by stating the equivalent problem as *X is typical given Y is typical* and using the formula. Both methods get full **2 marks** if done correctly. If deriving from first principles approach, the logical reasoning is seen even if not completely rigorous (like using ϵ and all)

Q6:

- a) **1 mark** for specifying $P(Y)$. **0.5 marks** if partially correct.
- b) Either representation using $H(Y) - H(Y|X)$ or $H(X) - H(X|Y)$ is fine. **Full marks** if both terms are correctly calculated.
- c) We understand that a closed form expression is not possible. Evaluation will proceed from your calculated $I(X; Y)$ and the steps you followed. Marks will be awarded based on **correctness** and **effort**. Mistakes in the derivative calculation will get penalised and deductions will be based on their severity and number.
- d) Please look in the attached solutions. Although, not from the previous step, an intuitive solution is still possible for this question. **1 mark** if the correct interpretation is given.

Q7:

Marks are distributed as **0.5** for modified variance calculation, **1.5** for 1st iteration, **1** for 2nd iteration, **1** for 3rd iteration and **1** for the final power allocation. Marks will be suitably deducted on mistakes in intermediate steps depending on correctness of solution elsewhere.

Some have calculated in a different way, taking the best k channels and checking iteratively how much power can be allocated. As long as the method is clearly explained, full marks will be given.

Q8:

2 methods, one by series expansion and another by L'Hopital's rule, are acceptable. Common mistake is about missing the $\ln(2)$ term, which will lead to a deduction of 1 mark.

Q5. $y = g(x) = \begin{cases} 0 & \text{if } x=0 \\ 1 & \text{if } x=1 \text{ or } x=2 \end{cases}$

$\therefore P(y) = \begin{cases} \frac{1}{2} & y=0 \\ \frac{1}{2} & y=1 \end{cases}$

a) $H(X) = \frac{1}{2} \log_2 2 + 2 \times \frac{1}{4} \log_2 4$

$= 1.5 \text{ bits}$

$H(Y) = 2 \times \frac{1}{2} \log_2 2$

$= 1 \text{ bit}$

b) y^n is a ~~typ~~ strongly-typical seq. in Y

Given, such a y^n , we want to find ~~if~~ how many x^n s.t. $g(x_i) = y_i \forall i$ are $x^n \in A_\epsilon^{(n)}(X)$

This can be interpreted as strongly typical sequences in the r.v. $X|Y$.

$\therefore |A_\epsilon^{(n)}(X|Y)| \approx 2^{nH(X|Y)}$

$H(X|Y) + H(Y) = H(X, Y)$

Note that since ~~$Y = f(X)$~~ $Y = g(X)$

$\Rightarrow H(X, Y) = H(X)$

$\Rightarrow H(X|Y) = H(X) - H(Y)$

$= 0.5$

$\therefore |A_\epsilon^{(n)}(X|Y)| \approx 2^{n(0.5)}$

Method 2:

If y^n is strongly ϵ -typical in Y , then

$$l = \#S \in n(0.5 \pm \epsilon)$$

This follows from the fact that a strongly typical sequence has the fraction of each symbol nearly equal to its probability.

For the corresponding x^n strongly typical in X , such that $g(x_i^n) = y_i^n \forall i$

$$\#1 \text{ in } x^n = t \in (0.25 \pm \epsilon')n$$

$\therefore$ we are being given a y^n , we only have to care about the possible choices of S in y^n to either 1 or 2 in x^n .

$\therefore$ No. of such seq. in $x^n =$ No. of ways of placing t 1's in l positions (of S in y^n)

$$= \sum_{t=(0.25-\epsilon')n}^{(0.25+\epsilon')n} \binom{l}{t}$$

$$\simeq (1 + 2\epsilon'n) \cdot 2^{lH(t/l)} \quad \left(\begin{array}{l} 1 \text{ comes from} \\ \text{the limit of} \\ \epsilon' \rightarrow 0 \end{array} \right)$$

$$\in 2^{\log(1+2\epsilon'n)} \cdot 2^{n(0.5 \pm \epsilon)(H(1/2) \pm \delta)}$$

(for some const. $\delta > 0$, depending on ϵ, ϵ')

$$\subseteq 2^{n(0.5 \pm \epsilon)(1 \pm \delta)} + \log(1 + 2\epsilon'n)$$

$$\subseteq 2^{n(0.5 \pm \delta')}$$

(for some constant $\delta' > 0$ as a fⁿ of ϵ and δ)

(ignore $\log n$ term w.r.t. to n)

Q6. a) $P(X=0) = \pi$

$$\begin{aligned} P(Y=0) &= \sum_{x \in \mathcal{X}} P(Y=0|X=x) \cdot P(X=x) \\ &= (1-\alpha)\pi + 0 \cdot (1-\pi) \\ &= (1-\alpha)\pi \end{aligned}$$

Similarly, $P(Y=1) = (1-\beta)(1-\pi)$
 $P(Y=e) = \alpha\pi + \beta(1-\pi)$

b) $I(X;Y) = H(Y) - H(Y|X)$

$$\begin{aligned} H(Y) &= -\pi(1-\alpha) \log(\pi(1-\alpha)) - (1-\pi)(1-\beta) \log((1-\pi)(1-\beta)) \\ &\quad - (\alpha\pi + \beta(1-\pi)) \log(\alpha\pi + \beta(1-\pi)) \end{aligned}$$

$$\begin{aligned} H(Y|X) &= P(X=0) \cdot H(Y|X=0) + P(X=1) \cdot H(Y|X=1) \\ &= \pi H(\alpha) + (1-\pi) H(\beta) \end{aligned}$$

(once X is known, it is a simple 2-choice channel)

Alternatively, consider

$$I(X;Y) = H(X) - H(X|Y)$$

$$H(X) = H(\pi)$$

$$\begin{aligned} H(X|Y) &= P(Y=0) \cdot H(X|Y=0) + P(Y=1) \cdot H(X|Y=1) \\ &\quad + P(Y=e) H(X|Y=e) \end{aligned}$$

$\therefore Y=0$ only if $X=0$ and $Y=1$ only if $X=1$ then
 $H(X|Y=0) = 0 = H(X|Y=1)$ (no uncertainty)

$$\text{Now, } P(X=0|Y=e) = \frac{P(Y=e|X=0) \cdot P(X=0)}{P(Y=e)} \quad \left(\begin{array}{l} \text{Bayes'} \\ \text{Theorem} \end{array} \right)$$

$$= \frac{\alpha\pi}{\alpha\pi + (1-\pi)\beta}$$

$$\therefore H(X|Y) = (\alpha\pi + \beta(1-\pi)) H\left(\frac{\alpha\pi}{\alpha\pi + (1-\pi)\beta}\right)$$

$$\therefore I(X;Y) = H(\pi) - (\alpha\pi + \beta(1-\pi)) H\left(\frac{\alpha\pi}{\alpha\pi + (1-\pi)\beta}\right)$$

c) We can derive that $\frac{dH(x)}{dx} = \ln\left(\frac{1-x}{x}\right)$

(assume H has log in base e for simplicity)

$$C = \max_{\pi} I(X;Y)$$

$$\frac{dI(X;Y)}{d\pi} = \ln\left(\frac{1-\pi}{\pi}\right) - \left[(\alpha-\beta) H\left(\frac{\alpha\pi}{\alpha\pi + (1-\pi)\beta}\right) \right] - \left[(\alpha\pi + \beta(1-\pi)) \ln\left(\frac{\beta(1-\pi)}{\alpha\pi}\right) \left(\frac{\alpha\beta}{(\alpha\pi + (1-\pi)\beta)^2} \right) \right]$$

chain rule : deriv. of $H(\cdot)$ derivative of the term in $H(\cdot)$

$$\text{Let } \frac{\beta(1-\pi)}{\alpha\pi} = \gamma ; \frac{\alpha\beta}{\alpha\pi + (1-\pi)\beta} = \frac{\alpha\gamma + \beta}{\gamma + 1}$$

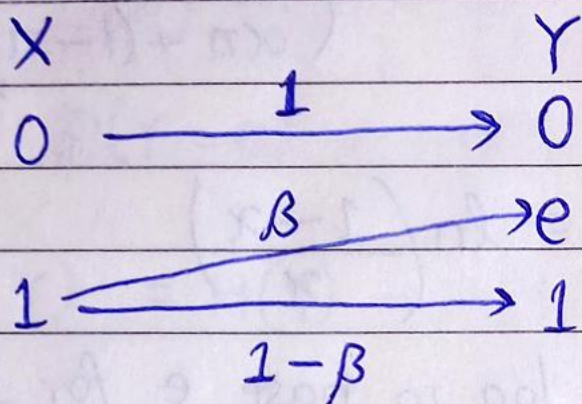
$$\text{Then } \frac{dI(X;Y)}{d\pi} = \ln\left(\frac{\alpha\gamma}{\beta}\right) - (\alpha-\beta) H\left(\frac{1}{1+\gamma}\right) - \ln(\gamma) \cdot \left(\frac{\alpha\gamma + \beta}{\gamma + 1}\right) = 0$$

$$\gamma \in (0, \infty)$$

for π^* and C

No closed form expression.

d) Although it is not clear how to interpret $\alpha=0$ from the prev. eqn, we can easily see from the channel characteristic.



Every symbol received in Y can be uniquely traced to either 0 or 1 in X .

Thus, $H(X|Y) = 0$

$\Rightarrow I(X; Y) = H(X) = H(\pi)$

$\therefore C = \max I(X; Y)$

$= 1$ when $\pi = \frac{1}{2}$ (uniform in X)

Q7. For the i^{th} subchannel, the channel eq's are

$$Y_i = |h_i| X_i + Z_i$$

$$\text{s.t. } Z_i \sim \mathcal{N}(0, \sigma^2) \quad \forall i \in \{1, 2, \dots, 8\}; \quad \sigma^2 = 8$$

$$\{|h_i|\}_{i=1}^8 = \{0.5, 1, 2, 4, 0.1, 0.2, 3, 5\}$$

The equivalent formulation can be obtained by dividing throughout by $|h_i|$

$$\Rightarrow Y_i' = X_i + Z_i'$$

with $Y_i' = \frac{Y_i}{|h_i|}, \quad Z_i' = \frac{Z_i}{|h_i|}$

$$\text{Thus, } Z_i' \sim \mathcal{N}\left(0, \frac{\sigma^2}{|h_i|^2}\right) \quad \text{for each } i$$

The power allocation problem is solved by the 'water-filling' algorithm such that

$$\sum_i P_i = P = 145$$

$$\lambda - \frac{\sigma^2}{|h_i|^2} > 0 \quad \forall \text{ subchannels } i \text{ where } P_i > 0$$

* P_i is power allocated to subchannel i
and $P_i \geq 0 \quad \forall i$

ALGORITHM STEPS:

$$\begin{aligned} \text{Iter.1} \quad \lambda^{(1)} &= \frac{P + \sum_i (\sigma^2 / |h_i|^2)}{8} \\ &= \frac{145 + 8 \sum_i (1 / |h_i|^2)}{8} \\ &= \frac{145 + 8 \times 130.4636}{8} \\ &= 148.5886 \end{aligned}$$

Now, assign $P_i = 0$ wherever $\lambda^{(1)} < \frac{\sigma^2}{|h_i|^2}$

$$\text{For } |h_5| = 0.1, \quad \frac{\sigma^2}{|h_5|^2} = 800 > \lambda^{(1)}$$

$$|h_6| = 0.2, \quad \frac{\sigma^2}{|h_6|^2} = 200 > \lambda^{(1)}$$

$$|h_7| = 0.5, \quad \frac{\sigma^2}{|h_7|^2} = 32 < \lambda^{(1)}$$

$\therefore$ Subchannels 5, 6 have $P_i = 0$. All others are fine.

Iter.2 We have subchannels $S = \{1, 2, 3, 4, 7, 8\}$ left

$$\begin{aligned} \lambda^{(2)} &= \frac{P + \sum_{i \in S} \sigma^2 / |h_i|^2}{|S|} \\ &= \frac{145 + 8 \times \sum_{i \in S} 1 / |h_i|^2}{6} \\ &= \frac{145 + 8 \times 5.4636}{6} \\ &= 31.45148 \end{aligned}$$

Again, set $P_i = 0$ where $\lambda^{(2)} < \frac{\sigma^2}{|h_i|^2}$

$$\text{For } |h_1| = 0.5; \frac{\sigma^2}{|h_1|^2} = 32 > \lambda^{(2)}$$

$$|h_2| = 1; \frac{\sigma^2}{|h_2|^2} = 8 < \lambda^{(2)}$$

thus, all others will have $\frac{\sigma^2}{|h_i|^2} < \lambda^{(2)}$

$\therefore P_1 = 0$; subchannel 1 is removed

Iter 3. Now $S_1 = \{2, 3, 4, 7, 8\}$ for leftower subchannels.

$$\lambda^{(3)} = \frac{145 + 8 \times 1.4636}{5}$$

$$= 31.3417$$

$$\text{For } |h_2| = 1, \frac{\sigma^2}{|h_2|^2} = 8 < \lambda^{(3)}$$

Thus all channels now satisfy $\frac{\sigma^2}{|h_i|^2} < \lambda^{(3)}$

$\therefore$ Assign $P_i = \lambda^{(3)} - \frac{\sigma^2}{|h_i|^2}$ for $i \in S_1$

$\therefore$ Power allocation table:

Channel	1	2	3	4	5	6	7	8
P_i	0	23.34	29.34	30.84	0	0	30.45	31.02

$$Q.8. C(W, P) = W \log_2 \left(1 + \frac{P}{N_0 W} \right)$$

$$C(\infty, P) := \lim_{W \rightarrow \infty} C(W, P)$$

$$:= \lim_{W \rightarrow \infty} \frac{\log_2 \left(1 + \frac{P}{N_0 W} \right)}{\frac{1}{W}}$$

This is of the form of $\frac{0}{0}$ indeterminate.
Using L'Hôpital's rule

$$\Rightarrow C(\infty, P) = \lim_{W \rightarrow \infty} \underbrace{\left[\left(\frac{1}{\ln 2} \right) \left(\frac{1}{1 + \frac{P}{N_0 W}} \right) \cdot \left(\frac{-P}{N_0 W^2} \right) \right]}_{\text{derivative of numerator}} \underbrace{\left(\frac{1}{-\frac{1}{W^2}} \right)}_{\text{derivative of denominator}}$$

$$= \lim_{W \rightarrow \infty} \frac{1}{\ln 2} \left(\frac{1}{1 + \frac{P}{N_0 W}} \right) \cdot \frac{P}{N_0}$$

$$= \frac{1}{\ln 2} \cdot \frac{P}{N_0}$$

Alternate : Using series expansion of $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$

$$C(\infty, P) = \lim_{W \rightarrow \infty} \frac{W}{\ln 2} \left[\frac{P}{N_0 W} - \frac{\left(\frac{P}{N_0 W} \right)^2}{2} + \frac{\left(\frac{P}{N_0 W} \right)^3}{3} - \dots \right]$$

$$= \lim_{W \rightarrow \infty} \frac{1}{\ln 2} \left[\frac{P}{N_0} - \frac{P^2}{2N_0^2 W} + \frac{P^3}{3N_0^3 W^2} - \dots \right]$$

$$= \frac{1}{\ln 2} \cdot \frac{P}{N_0} \quad \left(\text{all other terms contain } \frac{1}{W} \rightarrow 0 \text{ as } W \rightarrow \infty \right)$$